

Vertex crossings and boundary regimes in a symmetric Markov multinomial model

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Abstract

Take a Markov chain on d states that starts uniformly, stays in its current state with probability p , and otherwise moves to one of the other $d - 1$ states at random. We count how often it visits each state and compare the probability of a given count vector with the probability of a vertex, where every visit is to the same state. Writing the law as a sum with positive terms, we get a Bessel approximation whose relative error is bounded for every positive count vector, even near the boundary of the simplex. From it we find the crossing points for balanced faces and for interior points, with explicit constants. When at least two coordinates are of order N , a small coordinate starts to matter once it has size $N/(\log N)^2$. Near a vertex, where one count b is much larger than the total $A = o(b)$ of the others, the crossing is given by a product of Laguerre polynomials up to a relative error $O_\ell(\sqrt{(A+1)/b})$, and this can be true even when the matching approximation to the probabilities fails. We also find the volumes of the superlevel sets, and a weak field under which every support size is the mode in turn. The exact law and the normal limit were known before, and we include them for reference.

1 Introduction

When p is close to 1 the chain rarely switches, so the most likely count vector is a vertex. As p decreases, at some point a vector in the interior becomes more likely instead. We want to know where that happens. Note that we compare the probability of one vector, not of a whole face, and this makes the answer depend a lot on coordinates that are small compared with N . We keep the number of states fixed but let the count vector approach a vertex at any rate.

This is the version of Kagey's Problem 131 [1] with d directions instead of 2. The binary case, including free and periodic boundaries, is in a companion paper [2], but we do not use it here and we prove every estimate we need. One way to picture the model is a board shaped like a simplex, where each bounce goes in one of d fixed directions and a switch goes to each other direction equally often. A board with different rules at its walls would give a different process.

There are three regimes. On a fixed face of the simplex, the balanced crossing is at $Nu = \log N - \frac{1}{2} \log \log N + O(1)$, with a constant that depends on the size of the support. A Bessel bound that holds for all positive count vectors then handles the case where some coordinates have size $N/(\log N)^2$. Finally, if one count is b and the remaining counts have total $A = o(b)$, a single polynomial equation gives the crossing with relative error $O_\ell(\sqrt{(A+1)/b})$, uniformly over their individual sizes. This holds even in a regime where the corresponding approximation to the probabilities is not uniform. For the modes, we find the volumes of the spatial superlevel sets and give an explicit weak field under which every support size occurs in turn.

Relation to established results. The law belongs to the theory of Markov multinomial distributions [3, 4], and the exact finite sum and the normal limit below are included for reference. We cite the Wang–Yang article through Wang and Tang, since we have not been able to obtain it. Brydges, van der Hofstad and König [5] give occupation densities for continuous-time chains on the visited faces of the simplex. The square-root quantity that appears below is also classical: if Q is the continuous-time generator of the complete graph with off-diagonal entries one, then its symmetric occupation rate is

$$I_Q(\alpha) = -\langle \sqrt{\alpha}, Q\sqrt{\alpha} \rangle = d - \left(\sum_i \sqrt{\alpha_i} \right)^2.$$

Hence the exponent $(\sum_i \sqrt{\alpha_i})^2 - 1$ is the difference between the rate at a vertex and the rate at α . We take the Bessel expansions from [6] and prove the rest ourselves. What we add is that the estimates hold uniformly when p depends on N , and what that says near the boundary and about the modes.

Organization. In Section 2 we write down the exact law, a few simple bounds, the covariance and the normal limit for fixed p . Section 3 has the face crossings and the balancing theorem. Sections 4 and 5 deal with small coordinates and the near-vertex law, and Section 6 with superlevel sets and weak fields. We do not find every global mode at finite N , and the number of states stays fixed. In fact, at finite N the mode does not have to be a vertex or a full center (Example 3.8).

2 The model and exact formulas

Fix $d \geq 2$ and let $(X_j)_{j \geq 1}$ be a Markov chain on $\{1, \dots, d\}$. The initial state is uniform. At each subsequent step the chain repeats its state with probability $p \in [0, 1]$ and moves to each other state with probability $r = (1 - p)/(d - 1)$. Put $q = 1 - p$ and $\sigma = p - r = (dp - 1)/(d - 1)$. Let $K_N = (K_{N,1}, \dots, K_{N,d})$ be the occupation vector over N visits, and write $P_N(k) = \mathbb{P}(K_N = k)$. For a word w of length $N \geq 1$ with j changes, its probability is $d^{-1}p^{N-1-j}r^j$. The vertex probability is $V_N = p^{N-1}/d$. For $p > 0$ put $u = r/p$, and for a positive vector $n = (n_1, \dots, n_k)$ with total N , write

$$S_n(u) = \frac{P_N(n_1, \dots, n_k, 0, \dots, 0)}{V_N} = \sum_j W_n(j) u^j,$$

where $W_n(j)$ is the number of words on the k states with these counts and j changes. Note that this ratio does not depend on d , only on the positive coordinates. Set $P_0(0, \dots, 0) = 1$ separately.

Proposition 2.1 (d directions). *Let $P_N(k_1, \dots, k_d)$ be the probability that direction i is taken k_i times. Put $\sigma = p - \frac{q}{d-1} = \frac{dp-1}{d-1}$ and $\mathcal{R} = \sum_{i=1}^d \frac{x_i t}{1 - \sigma x_i t}$. Then*

$$\sum_{N \geq 0} \sum_{k_1 + \dots + k_d = N} P_N(k_1, \dots, k_d) x_1^{k_1} \dots x_d^{k_d} t^N = 1 + \frac{\mathcal{R}/d}{1 - \frac{q}{d-1} \mathcal{R}}.$$

Proof. Let F_i be the generating function of words ending in state i , and $S = \sum_i F_i$. Conditioning on the last bounce, $F_i = x_i t (\frac{1}{d} + p F_i + \frac{q}{d-1} (S - F_i))$, that is $F_i(1 - \sigma x_i t) = x_i t (\frac{1}{d} + \frac{q}{d-1} S)$. Summing over i gives $S = \mathcal{R}(\frac{1}{d} + \frac{q}{d-1} S)$, and the generating function is $1 + S$. $\square$

For $d = 2$, setting $x_1 = x$ and $x_2 = 1$ gives the binary generating function studied in the companion [2].

Proposition 2.2 (finite bin probabilities in d directions). *Put $r = q/(d-1)$ and $\sigma = p - r$. For $N \geq 1$, let $k = (k_1, \dots, k_d)$ have nonnegative integer coordinates summing to N , and put $H = \{i : k_i > 0\}$. Sum over $m_i = 0$ for $i \notin H$ and $1 \leq m_i \leq k_i$ for $i \in H$, and set $M = \sum_i m_i$. Then*

$$P_N(k) = \frac{1}{d} \sum_m \frac{M!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in H} \binom{k_i - 1}{m_i - 1}.$$

All exponents are nonnegative, and $0^0 = 1$. Set $P_0(0, \dots, 0) = 1$ separately.

Proof. With $z_i = x_i t$, Proposition 2.1 becomes the formal power series

$$1 + \frac{1}{d} \sum_{M \geq 1} r^{M-1} \left(\sum_i \frac{z_i}{1 - \sigma z_i} \right)^M.$$

Expand by the multinomial theorem and use, for $k \geq m \geq 1$,

$$[z^k] \left(\frac{z}{1 - \sigma z} \right)^m = \binom{k-1}{m-1} \sigma^{k-m}.$$

For $m = 0$ only $k = 0$ contributes. Multiplying the coefficients proves the formula. $\square$

At $d = 3$ this is a triple sum, with the support convention covering edge bins. At $p = 1/d$, $\sigma = 0$ and only $M = N$ survives, giving the ordinary multinomial law. At $p = 1$, only $M = 1$ survives, giving mass $1/d$ at each Ne_i . The formula also holds at $p = 0$. For $\sigma < 0$ its terms can cancel, so for numerical work we use dynamic programming or exact arithmetic. This model assumes fixed global directions and uniform switching to the other directions, and a board with a different rule need not have this law.

2.1 Elementary bounds used below

For $x \geq 0$, the modified Bessel function used here is

$$I_1(x) = \sum_{j \geq 0} \frac{(x/2)^{2j+1}}{j!(j+1)!}.$$

Its coefficients give $I_1(x) \leq \sinh x \leq e^x$. We also use the standard expansion [6, Eq. 10.40.1]

$$I_1(x) = \frac{e^x}{\sqrt{2\pi x}} \left(1 - \frac{3}{8x} + O(x^{-2}) \right), \quad x \rightarrow \infty.$$

The Laguerre–Bessel comparison needed near the boundary is proved in Lemma 4.1.

Lemma 2.3 (clipped product deficit). *For any finite list of nonnegative real numbers $t_1, \dots, t_m$, with $(x)_+ = \max(x, 0)$,*

$$0 \leq 1 - \prod_{j=1}^m (1 - t_j)_+ \leq \sum_{j=1}^m t_j.$$

Proof. If some $t_j \geq 1$, the product is zero and the sum is at least one. Otherwise all factors lie in $[0, 1]$ and induction follows from $1 - (1 - a)(1 - b) = a + b - ab \leq a + b$. The empty product gives equality. $\square$

2.2 Covariance and the fixed-persistence normal limit

We record these formulas for comparison with the later sections. Note that the normal limit at fixed p says nothing about individual occupation probabilities when p depends on N .

Remark 2.4 (covariance and the tetrahedral case). For the board with d directions of Proposition 2.1 let K_a be the number of bounces in direction a . The direction chain has transition matrix $\sigma I + (1 - \sigma)\frac{1}{d}J$ with J the all-ones matrix, so the indicator of direction a at time i and of direction b at time $i + m$ have covariance $\sigma^m(\frac{\delta_{ab}}{d} - \frac{1}{d^2})$. Summing over the two time indices and evaluating the finite geometric sum gives

$$\text{Cov}(K_a, K_b) = \left(\frac{\delta_{ab}}{d} - \frac{1}{d^2}\right) \left(N \frac{1 + \sigma}{1 - \sigma} - \frac{2\sigma(1 - \sigma^N)}{(1 - \sigma)^2}\right), \quad \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

These formulas hold for $p < 1$, and at $p = 1$ the covariance is $N^2(\delta_{ab}/d - 1/d^2)$. The asymptotic multiplier of the multinomial covariance is $(d - 2 + dp)/(d(1 - p))$, which is $(1 + 3p)/(3(1 - p))$ for the tetrahedral board, $d = 3$.

Theorem 2.5 (simplex normal limit). *Let $d \geq 3$ and $0 \leq p < 1$, or $d = 2$ and $0 < p < 1$. Set*

$$\pi = (1/d, \dots, 1/d), \quad C = \frac{1}{d}I - \frac{1}{d^2}J, \quad c = \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

Then $(K_N - N\pi)/\sqrt{N}$ converges in distribution to $\mathcal{N}_d(0, cC)$. This Gaussian is supported on the sum-zero hyperplane and is nondegenerate there.

Proof. Let $\Pi = \mathbf{1}\pi$ and $T = \sigma I + (1 - \sigma)\Pi$. Fix a real vector a and put $f_i = a_i - d^{-1}\sum_j a_j$, so $\pi f = 0$ and $Tf = \sigma f$. For $D(t) = \text{diag}(e^{itf_i})$ and $B(t) = TD(t)$, stationarity gives

$$\mathbb{E} \exp \left(it \sum_{j=1}^N f(X_j) \right) = \pi D(t) B(t)^{N-1} \mathbf{1}.$$

At zero, $B(0) = T$ has the simple eigenvalue 1 and all other eigenvalues equal σ , with $|\sigma| < 1$. Analytic perturbation gives an eigenvalue $\lambda(t)$ and right eigenvector $h(t)$ with $\lambda(0) = 1$, $h(0) = \mathbf{1}$ and $\pi h(t) = 1$. The rest of the spectrum is uniformly exponentially small for small enough t . To see this, take a circle of radius $\eta < 1$ around the rest of the spectrum but not around $\lambda(t)$, and use the resolvent formula for matrix powers. Thus the characteristic function equals $b(t)\lambda(t)^{N-1} + O(\eta^N)$ with $b(t) \rightarrow 1$.

Differentiate $Bh = \lambda h$ at zero. Since $B'(0) = iT \text{diag}(f)$,

$$\lambda'(0) = 0, \quad (I - T)h'(0) = i\sigma f, \quad h'(0) = \frac{i\sigma}{1 - \sigma}f.$$

Differentiating again and multiplying by π gives

$$\lambda''(0) = \pi B''(0)\mathbf{1} + 2\pi B'(0)h'(0) = -\frac{1 + \sigma}{1 - \sigma}\pi(f^2) = -c\pi(f^2).$$

Hence $\lambda(t) = 1 - c\pi(f^2)t^2/2 + O(t^3)$. Substitute $t = s/\sqrt{N}$ to obtain the scalar normal characteristic function with variance $c\pi(f^2) = ca^\top Ca$. Lévy's theorem and the Cramér–Wold device give the result. $\square$

For $d = 3$ the first two centered coordinates have limiting covariance

$$\frac{1+3p}{27(1-p)} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix};$$

the third is minus their sum. Unlike the binary case, $p = 0$ is included for $d \geq 3$. At $d = 2, p = 0$ bounded alternation gives the zero limit after division by $\sqrt{N}$. At $p = 1$ the centered sequence on this scale is not tight.

3 Face-center thresholds on a simplex

We keep the symmetric d -direction model of Proposition 2.1. The initial direction is uniform, a repeat has probability p , and each particular change has probability $r = (1-p)/(d-1)$. Other ways of modeling a tetrahedral board ($d = 3$) give different laws, which we do not treat. For $2 \leq k \leq d$ and $N \geq k$, let $C_{N,k}(p)$ denote the probability of one *balanced face bin*, meaning that its k positive coordinates differ by at most one and its other coordinates are zero. All such bins have the same probability. Let $V_N(p) = p^{N-1}/d$ be the probability of one vertex bin, and put

$$u = \frac{1-p}{(d-1)p}, \quad a_k = \frac{1}{2} \log(4\pi) - \frac{k+1/2}{k-1} \log k.$$

The ratio $C_{N,k}/V_N$, viewed as a function of u , depends on k and N but not on the ambient number d of directions.

Theorem 3.1 (face-center crossings). *For every $2 \leq k \leq d$ and $N \geq k$, there is exactly one $p = p_{N,k}^{(d)} \in (0, 1)$ for which $C_{N,k}(p) = V_N(p)$. The face center is more likely below this persistence and less likely above it. The sequence $p_{N,k}^{(d)}$ is strictly increasing in N , and*

$$p_{k,k}^{(d)} = \frac{1}{1 + (d-1)(k!)^{-1/(k-1)}}.$$

Writing $L = \log N$ and $u_{N,k} = (1 - p_{N,k}^{(d)})/((d-1)p_{N,k}^{(d)})$, for fixed k we have

$$Nu_{N,k} = L - \frac{1}{2} \log L + a_k + \frac{\frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}}{L} + O_k\left(\frac{(\log L)^2}{L^2}\right). \quad (1)$$

Consequently, $N(1 - p_{N,k}^{(d)})$ equals $(d-1)$ times the right-hand side of (1), with the same error order for fixed d .

In particular, for the full center $k = d$,

$$N(1 - p_{N,d}^{(d)}) = (d-1) \log N - \frac{d-1}{2} \log \log N + c_d + o(1), \quad c_d = \frac{d-1}{2} \log(4\pi) - \frac{2d+1}{2} \log d.$$

For $d = 2$, $a_2 = \frac{1}{2} \log(\pi/8)$, and the expansion agrees with the binary one in [2]. The proof below does not use the binary case and treats every face dimension at once. Figure 1 shows the full-center crossing on the triangle at $N = 24$.

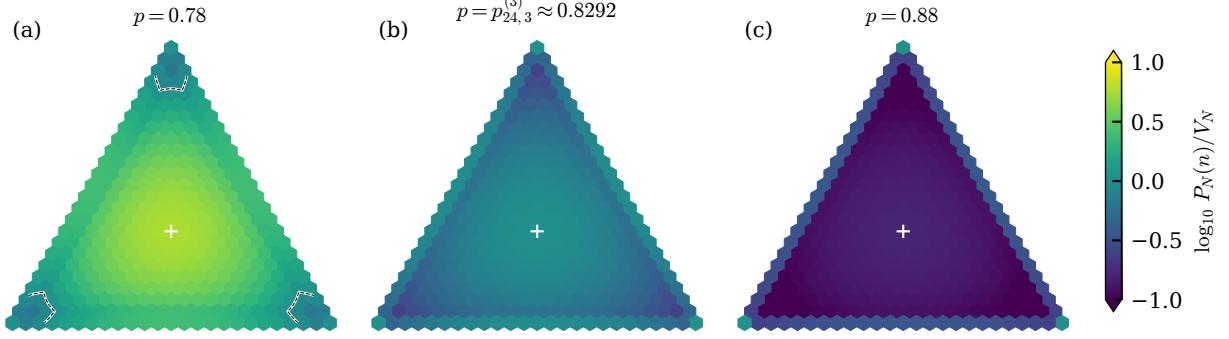


Figure 1: The ratio $P_N(n)/V_N$ for $d = 3$ and $N = 24$, one hexagon per bin, on a $\log_{10}$ scale. (a) At $p = 0.78$ only the 21 bins inside the dashed curves near the vertices are below the vertex height. (b) At the full-center crossing $p_{24,3}^{(3)} = 0.82921 \dots$ the center $(8, 8, 8)$ (cross) ties with the vertices and every other bin is below them. (c) At $p = 0.88$ every nonvertex bin is below the vertex height.

Theorem 3.2 (common critical window). *Fix d and put*

$$u_N(t) = \frac{\log N - \frac{1}{2} \log \log N + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)}.$$

Uniformly for t in any fixed compact interval, for each $2 \leq k \leq d$,

$$\frac{C_{N,k}(p_N(t))}{V_N(p_N(t))} \longrightarrow \exp((k-1)(t - a_k)). \quad (2)$$

Moreover $a_2 > a_3 > \dots$. Thus the face centers cross the vertex height at distinct constant offsets within a common window of persistence width $O(1/N)$.

It might seem strange that faces of every size cross in the same window. A word ending at a balanced vector on a face with k states has to visit all k of them. Each extra state needs its own runs, which means more switches, but those runs have about N places to go, and together this gives roughly a factor of e^{Nu}/N per state. So the ratio to a vertex is about $(e^{Nu}/N)^{k-1}$, which is 1 near $Nu \approx \log N$ no matter what k is. The size of the face only changes the constant a_k .

We first prove the exact assertions, and then prove a uniform local estimate from which both asymptotic theorems follow.

Proof of the exact assertions in Theorem 3.1. For a positive count vector $n = (n_1, \dots, n_k)$, let $W_n(j)$ be the number of words with those counts and exactly j changes. Its ratio to a vertex probability is

$$S_n(u) = \sum_j W_n(j) u^j.$$

To count $W_n(j)$, choose a proper word of $j+1$ run colors, meaning that successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. Thus $W_n(j)$ is a sum of products of these binomial coefficients. Every coefficient is nonnegative, and the first nonzero one is $W_n(k-1) = k!$. Hence S_n is strictly increasing from zero to infinity on $u \geq 0$ and crosses one exactly once. At $p = 0$, cycling through the k colors and truncating after N positions gives a word with balanced counts and no successive repeat, so its probability is positive while the vertex probability is zero. At $p = 1$ the reverse strict inequality holds.

Choose the balanced count vectors as N increases by incrementing one of their minimum coordinates. Each coefficient $W_n(j)$ is then nondecreasing. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2} (N - k).$$

Indeed, exactly one run color appears twice, and for each such color there are $(k+1)!/2 - k! = k!(k-1)/2$ proper arrangements of the multiset of run colors. This coefficient increases strictly with N , so the positive root decreases strictly with N . The substitution $p = 1/(1 + (d-1)u)$ proves the claimed monotonicity. If $N = k$, every coordinate equals one and $S_n(u) = k!u^{k-1}$, giving the initial value. Note also that every root satisfies $u \leq (k!)^{-1/(k-1)} < 1$, so all crossings lie in the positive refresh regime used below. $\square$

Lemma 3.3 (positive integral and discrete comparison). *For fixed $k \geq 2$, define*

$$F_k(x) = \sum_{m_1, \dots, m_k \geq 1} M! \prod_{i=1}^k \frac{x^{m_i}}{m_i! (m_i - 1)!} = \int_0^\infty e^{-t} [\sqrt{xt} I_1(2\sqrt{xt})]^k dt, \quad M = \sum_i m_i. \quad (3)$$

Let n be a balanced positive count vector of total N , put $h = N/k$, and, for $0 < u < 1$, put $x = hu/(1-u)$. With $\mathcal{D} = x d/dx$,

$$0 \leq 1 - \frac{S_n(u)}{(1-u)^{N-1} h^{1-k} F_k(x)/x} \leq \frac{\mathcal{D}^2 F_k(x)}{h F_k(x)} = O_k\left(\frac{(1+x)^2}{h}\right). \quad (4)$$

In particular the relative error is $O_k((\log N)^2/N)$ when $x = O(\log N)$.

Proof. The equality in (3) follows from the Bessel power series and Tonelli's theorem, using $M! = \int_0^\infty e^{-t} t^M dt$. The integral is finite for every positive x , since the bound $I_1(y) \leq e^y$ for $y \geq 0$ reduces its tail to a polynomial times $\exp(-t + 2k\sqrt{xt})$. Thus the positive series and all its termwise derivatives converge on compact subsets of $x > 0$.

Set $\sigma = p - r = p(1-u)$ and $v = r/\sigma = u/(1-u)$. Proposition 2.2, restricted to the positive coordinates, gives

$$S_n(u) = (1-u)^{N-1} \sum_{m_i \geq 1} M! v^{M-1} \prod_i \frac{\binom{n_i-1}{m_i-1}}{m_i!}. \quad (5)$$

Define

$$A_{i,m} = \frac{(m-1)!}{h^{m-1}} \binom{n_i-1}{m-1} = \prod_{j=1}^{m-1} \left(1 - \frac{h - n_i + j}{h}\right)_+.$$

Since $h - n_i > -1$, these are products of factors in $[0, 1]$, including beyond their finite support. The elementary product-deficit bound in Lemma 2.3 gives

$$0 \leq 1 - \prod_i A_{i,m_i} \leq \frac{1}{h} \sum_i \frac{m_i(m_i+1)}{2} \leq \frac{M^2}{h}.$$

Replacing the binomial factors in (5) by $h^{m_i-1}/(m_i-1)!$ gives the denominator in (4). Summing the deficit bound proves its first inequality and the displayed explicit upper bound.

To bound the quotient of derivatives, for $x \geq 1$ use $M^2 \leq 3x^2(1+1/x)^M$. Indeed, $\log(1+1/x) \geq 1/(2x)$ and $\sup_{y \geq 0} y^2 e^{-y/2} = 16/e^2 < 3$. Positivity of the coefficients gives

$$\frac{\mathcal{D}^2 F_k(x)}{F_k(x)} \leq 3x^2 \frac{F_k(x+1)}{F_k(x)} = O_k(x^2),$$

where the last step follows from Lemma 3.4 below. For $0 < x \leq 1$, the quotient extends continuously to $x = 0$ with value k^2 , since $F_k(x) = k!x^k + O_k(x^{k+1})$. This proves the bound. $\square$

Lemma 3.4 (Laplace expansion). *For each fixed $k \geq 2$, as $x \rightarrow \infty$,*

$$F_k(x) = \frac{k^{k/2+1}}{(4\pi)^{(k-1)/2}} x^{(k+1)/2} e^{k^2 x} \left(1 - \frac{k-1}{8kx} + O_k(x^{-2}) \right). \quad (6)$$

Proof. In (3) set $w = \sqrt{t}$ and $a = k\sqrt{x}$. The exponent becomes $-w^2 + 2k\sqrt{x}w = k^2x - (w-a)^2$. First restrict to $|w-a| \leq \sqrt{x}/2$. There $2\sqrt{x}w$ is bounded below by $(2k-1)x$, so the standard Bessel expansion [6, Eq. 10.40.1] applies uniformly and gives

$$[\sqrt{x}w I_1(2\sqrt{x}w)]^k = \frac{x^{k/4} w^{k/2} e^{2k\sqrt{x}w}}{(4\pi)^{k/2}} \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2}) \right).$$

On the complementary set, the elementary exponential bound for I_1 dominates the integrand by $2x^{k/2} w^{k+1} e^{k^2x - (w-a)^2}$. Gaussian tail estimates therefore show that its integral is at most $e^{k^2x - cx}$ times a fixed power of x , for some $c > 0$. It is exponentially smaller than the main term.

Put $\alpha = k/2 + 1$. Apart from that negligible tail, the integral is

$$\frac{2x^{k/4} e^{k^2x}}{(4\pi)^{k/2}} \int_{|w-a| \leq \sqrt{x}/2} e^{-(w-a)^2} w^\alpha \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2}) \right) dw.$$

Now Taylor expand w^α about a . On this symmetric interval the odd terms integrate to zero. The quadratic term gives a relative correction of $\alpha(\alpha-1)/(4a^2)$, and the fourth-order remainder is $O_k(a^{-4})$, since the interval stays a fixed fraction of a away from zero and the Gaussian moments are finite. If we extend the Gaussian integrals to the whole line, they change by an exponentially small amount. For the term involving $w^{\alpha-1}/\sqrt{x}$, keeping its constant term suffices, because its quadratic correction has relative order x^{-2} . Since $\int e^{-y^2} dy = \sqrt{\pi}$ and $\int y^2 e^{-y^2} dy = \sqrt{\pi}/2$, the total relative correction is

$$\frac{\alpha(\alpha-1)}{4k^2x} - \frac{3}{16x} = -\frac{k-1}{8kx}.$$

The leading constant simplifies to the one in (6), completing the proof. $\square$

Proof of the asymptotic assertions in Theorems 3.1 and 3.2. Set $s = k-1$, $z = Nu$, and $D_k = k^{k+1/2}/(4\pi)^{(k-1)/2}$. The preceding lemmas imply, uniformly on $c \log N \leq z \leq C \log N$ for fixed $0 < c < C$,

$$S_n(z/N) = D_k \frac{z^{s/2} e^{sz}}{N^s} \left(1 - \frac{s}{8z} + O_{k,c,C}(z^{-2} + z^2/N) \right). \quad (7)$$

Indeed, $x = z/(k(1-u))$, and

$$k^2x + (N-1) \log(1-u) - sz = O_k(z^2/N).$$

The prefactor changes by $1 + O_k(z/N)$ when we replace x by z/k , and rounding of the balanced coordinates is covered by (4).

At $z = (1 - \varepsilon) \log N$ the ratio in (7) tends to zero, whereas at $z = (1 + \varepsilon) \log N$ it tends to infinity. The unique root therefore has $z / \log N \rightarrow 1$. Taking logarithms at the root and dividing by s gives

$$z + \frac{1}{2} \log z = L + a_k + \frac{1}{8z} + O_k(z^{-2} + z^2/N).$$

For an explicit inversion, put

$$d_L = -\frac{1}{2} \log L + a_k, \quad e_L = \frac{1}{8} - \frac{1}{2} d_L, \quad z_* = L + d_L + e_L/L.$$

Taylor expansion gives $z_* + \frac{1}{2} \log z_* - 1/(8z_*) = L + a_k + O_k((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2} \log z - 1/(8z)$ is at least one for $z > 0$. The preceding root equation, with $z \sim L$, therefore gives $z - z_* = O_k((\log L)^2/L^2)$, proving (1). The conversion to physical persistence uses $N(1 - p) = (d - 1)z/(1 + (d - 1)z/N) = (d - 1)z + O_d(z^2/N)$. Substituting $z = L - \frac{1}{2} \log L + t$ in (7) proves (2), uniformly on compact t intervals.

For the ordering of constants, extend $g(x) = (x + 1/2) \log x / (x - 1)$ to $x \geq 2$. The sign of $g'(x)$ is the sign of $H(x) = 2x^2 - x - 1 - 3x \log x$. Here $H(2) = 5 - 6 \log 2 > 0$, and $H'(x) = 4x - 4 - 3 \log x > 0$ for $x \geq 2$, since its derivative is $4 - 3/x > 0$ and $H'(2) > 0$. Thus g increases strictly, so a_k decreases strictly. $\square$

3.1 Thresholds throughout the interior of a face

The same argument gives a threshold at every point in the interior of a face, not just at its center. The constants below are evaluated at the actual fractions $\alpha_i = n_i/N$, since we do not assume that the count vector converges at any rate.

Theorem 3.5 (interior composition thresholds). *Fix $k \geq 2$ and $\varepsilon > 0$. Let $n_1 + \dots + n_k = N$, with $\alpha_i = n_i/N \geq \varepsilon$, and define*

$$A = \sum_{i=1}^k \sqrt{\alpha_i}, \quad g = A^2 - 1, \quad s = k - 1, \quad D(\alpha) = \frac{A^{k/2+1} (\prod_i \alpha_i)^{-3/4}}{(4\pi)^{s/2}}.$$

Uniformly for these count vectors and $c \log N \leq z \leq C \log N$, with fixed $0 < c < C$,

$$S_n(z/N) = D(\alpha) \frac{z^{s/2} e^{gz}}{N^s} (1 + O_{k,\varepsilon,c,C}(z^{-1} + z^2/N)). \quad (8)$$

Let $u_n > 0$ be the unique root of $S_n(u) = 1$. Writing $L = \log N$,

$$Nu_n = \frac{s}{g} L - \frac{s}{2g} \log L - \frac{\log D(\alpha) + (s/2) \log(s/g)}{g} + O_{k,\varepsilon} \left(\frac{\log L}{L} + \frac{L^2}{N} \right). \quad (9)$$

These estimates are uniform on every compact subset of the relative interior of the face. For a fixed target α and integer counts $n_i = N\alpha_i + O(1)$, the constants may equivalently be evaluated at that target, without changing the error order.

Proof. The proper-run polynomial used above has nonnegative coefficients and first coefficient $k!u^{k-1}$ for every positive count vector, which gives the unique positive root. To obtain the approximation, put $v = u/(1 - u)$ and $y = Nv$, and introduce

$$G_\alpha(y) = \int_0^\infty e^{-t} \prod_i [\sqrt{\alpha_i y t} I_1(2\sqrt{\alpha_i y t})] dt.$$

Replacing each binomial coefficient in (5) by $n_i^{m_i-1}/(m_i-1)!$ gives

$$S_n(u) = \frac{(1-u)^{N-1}}{v \prod_i n_i} G_\alpha(y) (1 + O_{k,\varepsilon}((1+y)^2/N)). \quad (*)$$

To see that the error is uniform, note that the falling-factorial deficit is bounded by $\sum_i m_i(m_i-1)/(2n_i) \leq M^2/(2\varepsilon N)$. The positive-series argument in Lemma 3.3 bounds the weighted second moment of M by a constant times $(1+y)^2$. Uniformity follows from the uniform saddle expansion below, which bounds $G_\alpha(y+1)/G_\alpha(y)$ on the compact set $\{\alpha_i \geq \varepsilon, \sum_i \alpha_i = 1\}$. For bounded y , continuity and the positive leading coefficient give the same conclusion.

Set $w = \sqrt{t}$. The exponential factor of the integrand is

$$\exp(-w^2 + 2A\sqrt{y}w) = \exp(A^2y - (w - A\sqrt{y})^2).$$

Every α_i stays bounded away from zero, so the Bessel expansions, Gaussian tail estimates, and Taylor remainders in Lemma 3.4 are uniform here. Keeping their leading terms yields

$$G_\alpha(y) = \frac{A^{k/2+1}(\prod_i \alpha_i)^{1/4}}{(4\pi)^{s/2}} y^{(k+1)/2} e^{A^2y} (1 + O_{k,\varepsilon}(y^{-1})).$$

Substitute this into (*) and use $v \prod_i n_i = yN^{k-1} \prod_i \alpha_i$. For $u = z/N$ in the stated range, $y = z + O(z^2/N)$ and $(N-1)\log(1-u) = -z + O(z^2/N)$. This proves (8) with the specified constant.

On the compact set of compositions, both g and $D(\alpha)$ have positive lower and finite upper bounds. Testing the ratio at suitable constant multiples of $\log N$ brackets the unique root uniformly. Taking logarithms there gives

$$gz + \frac{s}{2} \log z = sL - \log D(\alpha) + O_{k,\varepsilon}(z^{-1} + z^2/N).$$

First $z = (s/g)L + O_{k,\varepsilon}(\log L)$; one substitution then gives (9). The derivative of its left-hand side is at least g , bounded uniformly away from zero, so the stated error is preserved on inversion. Finally, the functions of α appearing in the formula have bounded derivatives on a slightly smaller compact set, so changing α by $O(1/N)$ changes the expression by $O(L/N)$, which is absorbed by its error term. $\square$

At a balanced composition, $A = \sqrt{k}$, $g = k-1$, and $D(\alpha) = D_k$, recovering the preceding center formula through its constant term. For $k=2$, $g = 2\sqrt{\alpha_1\alpha_2}$, recovering the binary bulk scale. Cauchy–Schwarz gives $g \leq k-1$, with equality only at the balanced composition, so the leading threshold coefficient s/g is minimized exactly at the center of each face.

3.2 Balanced bins and global modes

The idea is that once we fix how many runs each state gets, splitting the visits evenly gives the most ways to choose the run lengths. The log-concavity of B_n in the proof below makes this precise.

Theorem 3.6 (balancing within a support). *For $1/d \leq p < 1$, among bins with a prescribed set of positive coordinates and total N , probability is maximized precisely by the balanced count vectors. More precisely, transferring one unit from a coordinate b to another coordinate a , where $b \geq a+2$ and $a \geq 1$, strictly increases the bin probability.*

Proof. At $p = 1/d$ this follows from the ordinary multinomial formula. For $1/d < p < 1$, put $\sigma = p - r > 0$ and $v = r/\sigma > 0$. The positive refresh sum has the integral form

$$P_N(n) = \frac{\sigma^{N-1}}{dv} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt, \quad B_n(y) = \sum_{m=1}^n \binom{n-1}{m-1} \frac{y^m}{m!}. \quad (10)$$

For each $y > 0$, the sequence $B_n(y)$, $n \geq 1$, is strictly log-concave. This is a case of the fact that the binomial transform preserves log-concavity, and we give a direct proof. Write

$$\frac{B_n(y)}{y} = b_{n-1}, \quad b_j = \sum_{i=0}^j \binom{j}{i} a_i, \quad a_i = \frac{y^i}{(i+1)!}, \quad c_j = \sum_{i=0}^j \binom{j}{i} a_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the expectation of the strictly decreasing sequence $a_{i+1}/a_i = y/(i+2)$ under weights proportional to $\binom{j}{i} a_i$. When j increases to $j+1$, the likelihood ratio of the new and old unnormalized weights on the old support is $(j+1)/(j+1-i)$, which increases in i , and a new positive weight appears at $i = j+1$. Thus these weights shift strictly toward larger indices, so the expectation of the strictly decreasing sequence decreases strictly. To check this last step, cross-multiply the two expectations and pair the terms with indices $i < \ell$. Hence $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j , proving the asserted strict log-concavity.

It follows that $B_{a+1}(y)B_{b-1}(y) > B_a(y)B_b(y)$ whenever $b \geq a+2$. Apply this inequality pointwise for $y = vt > 0$ in (10) and integrate. Repeated transfers give the conclusion. $\square$

Corollary 3.7 (global modes in the critical window). *Fix d . In the window of Theorem 3.2, if $t < a_d$, then for all sufficiently large N the only global maxima are the d vertices. If $t > a_d$, then the only global maxima are the fully supported balanced bins. These assertions are uniform on compact sets of t separated from a_d . At the exact full-center/vertex crossing, for all sufficiently large N the global maxima are exactly the vertices and the fully supported balanced bins, with both classes present.*

Proof. Theorem 3.6 reduces the possible modes to balanced face bins of support sizes $1, \dots, d$. For $2 \leq k \leq d$ their limiting log ratios to a vertex are $(k-1)(t-a_k)$. If $t < a_d$, these are all negative. If $t > a_d$, the full-center ratio exceeds the vertex ratio, and for $2 \leq k < d$ its limiting log ratio exceeds the k -face one by

$$(d-k)(t-a_d) + (k-1)(a_k-a_d) > 0.$$

Uniformity in Theorem 3.2 gives both assertions. At the exact full-center crossing, its window parameter tends to a_d by Theorem 3.1, and every proper face-center ratio then has a limit strictly less than one. The same reduction proves the final claim. $\square$

Example 3.8 (an intermediate face can be a finite-size mode). Corollary 3.7 does not hold for every N . Take $d = 3$, $N = 4$, and $p = 10/17$, so $u = 7/20$. For an edge center and a full center respectively,

$$\frac{P_4(2, 2, 0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \quad \frac{P_4(2, 1, 1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

By Theorem 3.6, the three edge centers are the only global modes. So at finite N the modes can lie on an intermediate face.

4 A uniform simplex estimate and a boundary crossover

The compact-interior assumption in Theorem 3.5 excludes bins approaching a lower-dimensional face. We now remove that assumption from the Bessel estimate and study the transition in which some positive coordinates become small. The transition below needs at least two coordinates of order N . With only one such coordinate the persistence scale is different, and we treat that case at the end of this section and in Section 5.

Put

$$\Phi(x) = \sum_{j \geq 0} \frac{x^j}{j!(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}}, \quad \Phi(0) = 1.$$

As before, $S_n(u)$ is the probability of the positive count vector n divided by the probability of one vertex, and $u = r/p$. None of the statements depend on d once the support is fixed.

Lemma 4.1 (a uniform Laguerre–Bessel inequality). *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\Phi(aw)} \leq \frac{w}{2}, \quad B_a(w) = \sum_{m=1}^a \binom{a-1}{m-1} \frac{w^m}{m!}. \quad (11)$$

Proof. The coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\prod_{h=1}^j (1 - h/a)_+$. Its deficit from one lies between zero and $j(j+1)/(2a)$, including for $j \geq a$. Consequently

$$0 \leq \Phi(aw) - \frac{B_a(w)}{w} \leq \frac{1}{2a} \sum_{j \geq 0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2} \Phi(aw).$$

□

Theorem 4.2 (a uniform Bessel majorant on every simplex stratum). *Let $n_1, \dots, n_k$ be arbitrary positive integers with sum N , and let $0 < u < 1$, $v = u/(1-u)$. Define*

$$\mathcal{Q}_n(u) = (1-u)^{N-1} v^{k-1} \int_0^\infty e^{-t} t^k \prod_{i=1}^k \Phi(n_i v t) dt.$$

Then, with no lower bound on any n_i/N ,

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} \leq k(k+1)v + \frac{k}{2}v^2 \left(\sum_i \sqrt{n_i} \right)^2 \leq k(k+1)v + \frac{k^2}{2}Nv^2. \quad (12)$$

Thus this approximation has relative error $O_k((\log N)^2/N)$ whenever $u = O(\log N/N)$, uniformly over every positive composition of N .

Proof. The positive refresh formula gives

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt.$$

Applying Lemma 4.1 to each factor, and then the product-deficit inequality, bounds the relative deficit by $(kv/2)\mathbb{E}[T]$, where T has density proportional to $e^{-t}t^k \prod_i \Phi(n_i vt)$. For the probability weights proportional to $x^j/[j!(j+1)!]$, the identity $\mathbb{E}[J(J+1)] = x$ implies

$$x \frac{\Phi'(x)}{\Phi(x)} = \mathbb{E}[J] \leq \sqrt{x}.$$

Integration by parts therefore gives

$$\mathbb{E}[T] = k + 1 + \sum_i \mathbb{E} \left[\frac{n_i v T \Phi'(n_i v T)}{\Phi(n_i v T)} \right] \leq k + 1 + \sqrt{v} \left(\sum_i \sqrt{n_i} \right) \sqrt{\mathbb{E}[T]}.$$

The boundary terms vanish, since near zero the density is $O(t^k)$, while $\Phi(x) \leq e^{2\sqrt{x}}$ bounds the tail by a polynomial times $\exp(-t + 2\sqrt{vt} \sum_i \sqrt{n_i})$. This also proves finiteness of the integral and of every moment used above. Applying $2ab \leq a^2 + b^2$ to the final term yields $\mathbb{E}[T] \leq 2(k+1) + v(\sum_i \sqrt{n_i})^2$. The claimed estimates follow, the last one by Cauchy–Schwarz. $\square$

4.1 Coordinates of order $N/(\log N)^2$

A small coordinate a adds its own runs. Each run needs about two switches, so it gives a factor u^2 , and it can go in about N places. So what matters is aNu^2 , just like bu^2 in the binary edge law. At the crossing u is about $(\log N)/N$, and then aNu^2 is about $a(\log N)^2/N$, which is why coordinates of this size start to matter.

Fix integers $r \geq 2$ and $\ell \geq 1$, and put $k = r + \ell$. Consider r large counts $n_1, \dots, n_r$ and ℓ further positive counts $a_1, \dots, a_\ell$. Write

$$N = \sum_i n_i + \sum_j a_j, \quad M = \sum_i n_i, \quad L = \log N, \quad \beta_i = \frac{n_i}{M}, \quad \rho_j = \frac{a_j L^2}{N}.$$

The β_i and ρ_j below always refer to these actual counts. Define

$$A = \sum_{i=1}^r \sqrt{\beta_i}, \quad g = A^2 - 1, \quad K = r + \ell - 1, \quad H = \frac{r-1}{2} + 2\ell, \quad c = \frac{K}{g},$$

and

$$D_r(\beta) = \frac{A^{r/2+1} (\prod_i \beta_i)^{-3/4}}{(4\pi)^{(r-1)/2}}.$$

Theorem 4.3 (a crossover between neighboring simplex strata). *Fix $\varepsilon > 0$ and $R < \infty$. Assume $\beta_i \geq \varepsilon$ and $0 < \rho_j \leq R$. Uniformly for such count vectors and $c_0 L \leq z \leq C_0 L$, where $0 < c_0 < C_0 < \infty$ are fixed,*

$$S_{(n,a)}(z/N) = D_r(\beta) A^{2\ell} \frac{z^H e^{gz}}{N^K} \prod_{j=1}^{\ell} \Phi\left(\frac{A^2 a_j z^2}{N}\right) \left(1 + O\left(L^{-1} + \frac{L^2}{N}\right)\right). \quad (13)$$

The unique positive solution $u_{n,a}$ of $S_{(n,a)}(u) = 1$ consequently satisfies

$$\begin{aligned} Nu_{n,a} = cL - \frac{H}{g} \log L - \frac{1}{g} \left\{ \log(D_r(\beta) A^{2\ell}) + H \log c + \sum_{j=1}^{\ell} \log \Phi(A^2 c^2 \rho_j) \right\} \\ + O\left(\frac{\log L}{L} + \frac{L^2}{N}\right). \end{aligned} \quad (14)$$

The constants in the errors depend only on the displayed fixed parameters.

Proof. Let $v = u/(1 - u)$ and $Y = Mv$. By Theorem 4.2, with relative error $O_k(v + Nv^2)$, the exact ratio equals

$$\frac{(1 - u)^{N-1} v^{\ell-1}}{\prod_{i=1}^r n_i} \int_0^\infty e^{-t} t^\ell \prod_{i=1}^r \left[\sqrt{\beta_i Y t} I_1(2\sqrt{\beta_i Y t}) \right] \prod_{j=1}^\ell \Phi(a_j v t) dt. \quad (15)$$

In the stated regime $Y \asymp L$ and all $a_j v Y$ are bounded. Set $w = \sqrt{t}$. The exponential from the r large-coordinate factors is $\exp(A^2 Y - (w - A\sqrt{Y})^2)$. On $|w - A\sqrt{Y}| \leq \sqrt{Y}/2$, the large-coordinate Bessel arguments are bounded below by a positive constant times Y . Their usual large-argument expansions are uniform, with relative error $O(1/Y)$. The remaining smooth factor, apart from constants and powers of Y , is

$$w^{r/2+1+2\ell} \prod_j \Phi(a_j v w^2).$$

Divide this factor by its value at $w = A\sqrt{Y}$. On the same interval its second derivative is $O(1/Y)$, uniformly, because $w/\sqrt{Y}$ lies in a fixed compact subset of $(0, \infty)$, the quantities $a_j v Y$ lie in a fixed compact interval, and $\Phi \geq 1$ there. Taylor's theorem, followed by integration against the centered Gaussian, eliminates the linear term and gives relative error $O(1/Y)$.

The part outside this interval is exponentially small, uniformly. The bounds $I_1(x) \leq e^x$ and $\Phi(x) \leq e^{2\sqrt{x}}$ give a polynomial times

$$\exp \left(A^2 Y - (w - A\sqrt{Y})^2 + 2w \sum_j \sqrt{a_j v} \right).$$

Here $\sum_j \sqrt{a_j v} = O(Y^{-1/2})$, so completing the square leaves a Gaussian centered only $O(Y^{-1/2})$ away from $A\sqrt{Y}$. On the excluded region its integral is exponentially small relative to the main term, even after the polynomial factors are included. The same argument also justifies extending the central Gaussian integrals to the full real line.

Evaluation of the leading Gaussian integral in (15) now gives

$$S_{(n,a)}(u) = D_r(\beta) A^{2\ell} \frac{Y^H}{M^K} e^{A^2 Y + (N-1) \log(1-u)} \prod_j \Phi(A^2 a_j M v^2) \\ \times (1 + O(Y^{-1} + v + Nv^2)).$$

Since $\sum_j a_j/N = O(L^{-2})$, at $u = z/N$ we have

$$Y = z + O(L^{-1} + L^2/N), \quad A^2 Y + (N-1) \log(1-u) = gz + O(L^{-1} + L^2/N).$$

Replacing Y by z , M by N , and $A^2 a_j M v^2$ by $A^2 a_j z^2/N$ in the other factors changes the expression by a relative $O(L^{-2} + L/N)$, because the arguments of Φ stay bounded and $\Phi \geq 1$. This proves (13).

The proper-run polynomial has nonnegative coefficients, zero constant coefficient, and positive coefficient $k!$ at u^{k-1} , so the crossing is unique. On the compact set of permitted β , both g and $D_r(\beta)$ are bounded above and bounded away from zero. Testing (13) at fixed sufficiently small and large multiples of L brackets the root uniformly. Its logarithm gives

$$gz + H \log z = KL - \log(D_r(\beta) A^{2\ell}) - \sum_j \log \Phi(A^2 \rho_j (z/L)^2) + O(L^{-1} + L^2/N).$$

First $z = cL + O(\log L)$, so the sum of logarithms on the right differs from its value at $z = cL$ by $O(\log L/L)$. Substituting this estimate into $\log z$ proves (14). Inverting keeps the error because the derivative of $gz + H \log z$ is at least $g > 0$. $\square$

Corollary 4.4 (invisibility of subcritical rare counts). *Under the hypotheses of Theorem 4.3, if all $a_j = o(N/(\log N)^2)$, the sum involving Φ in (14) is $o(1)$. Thus, at fixed actual large composition β , the rare counts do not affect the expansion through its constant term. Their number ℓ still changes both its leading and its logarithmic coefficients.*

The scale $N/(\log N)^2$ applies near a face with at least two coordinates of order N . It does not apply at a binary endpoint, where only one coordinate is large and $g = 0$. That case is treated below and in Section 5. Theorem 4.3 gives no error bound when some ρ_j is unbounded.

Lemma 4.5 (exact insertion of a singleton color). *Let n have total size M and let $\mathcal{D} = u d/du$. Then*

$$S_{(n,1)}(u) = [2u + (M-1)u^2]S_n(u) + (u - u^2)\mathcal{D}S_n(u).$$

Proof. Insert the new color into a word of length M with j changes. The two end positions contribute a factor $2u$, the j change gaps contribute ju , and the $M-1-j$ repeat gaps contribute $(M-1-j)u^2$. Removing the unique new letter reverses this construction. Summing the resulting identity over all words proves the formula. $\square$

The singleton identity gives an exact finite-size check of the factor $A^2 z^2/N$ that one very small coordinate contributes in (13).

4.2 Fixed rare counts near a vertex

With only one large coordinate, the limit is a finite polynomial instead. In this result the rare counts are fixed, and the large coordinate b , not the total count, is the asymptotic parameter.

Theorem 4.6 (a multicolor Laguerre limit at a vertex). *Fix $\ell \geq 1$ and positive integers $a_1, \dots, a_\ell$. Define*

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad Q(y) = \sum_{i < j} B'_{a_i}(y) B'_{a_j}(y) \prod_{h \neq i,j} B_{a_h}(y).$$

The sum defining Q is zero when $\ell = 1$. Uniformly on compact subsets of $y > 0$, as $b \rightarrow \infty$,

$$S_{(b,a)}(\sqrt{y/b}) = H(y) + \frac{2\sqrt{y}}{\sqrt{b}}(H'(y) + yQ(y)) + O_a(b^{-1}). \quad (16)$$

Let $y_a > 0$ be the unique root of $H(y_a) = 1$, and put $c_a = \sqrt{y_a}$. The unique endpoint-crossing odds satisfy

$$u_{b,a} = \frac{c_a}{\sqrt{b}} - \frac{1 + y_a Q(y_a)/H'(y_a)}{b} + O_a(b^{-3/2}). \quad (17)$$

For one rare color this recovers the binary fixed-edge correction $-1/b$. For at least two rare colors the coefficient of $1/b$ is strictly less than -1 .

Proof. Fix rare-run counts $m_i \in \{1, \dots, a_i\}$, put $m = \sum_i m_i$, and let q be the number of runs of the large color. The rare run lengths can be chosen in $\prod_i \binom{a_i-1}{m_i-1}$ ways. Since successive large-color runs must be separated, $q \leq m+1$, and the large run lengths have $\binom{b-1}{q-1}$ choices. At $u = \sqrt{y/b}$, configurations with these run counts contribute $O_a(b^{(q-m-1)/2})$. There are finitely many possible m_i, q , independently of b , so only $q = m+1$ and $q = m$ affect the expansion through order $b^{-1/2}$.

If $q = m+1$, the run word starts and ends in the large color and alternates between large and rare colors. There are $m!/\prod_i m_i!$ rare orders. Using $\binom{b-1}{m} = b^m/m! + O_a(b^{m-1})$ and summing gives $H(y)$, with an $O_a(b^{-1})$ error.

If $q = m$, there are two possibilities. One endpoint is rare and the run word otherwise alternates, giving $2m!/\prod_i m_i!$ orders. These terms sum to $2\sqrt{y}H'(y)/\sqrt{b}$. Otherwise both endpoints are large and exactly one pair of consecutive rare runs is present. The members of that pair must have different colors. For an ordered pair of colors i, j , contracting the marked adjacent pair shows that there are $(m-1)!/[(m_i-1)!(m_j-1)!\prod_{h \neq i,j} m_h!]$ orders. Summing over the two orientations and all $i < j$ gives $2y^{3/2}Q(y)/\sqrt{b}$. Terms with $q \leq m-1$ and the next terms of the binomial expansions are $O_a(b^{-1})$, uniformly for y in a positive compact set. This proves (16).

The polynomial H has positive coefficients, tends to zero at zero, and grows to infinity, so y_a exists uniquely and $H'(y_a) > 0$. The same positivity argument gives the unique crossing for each b . First bracket it using (16) to obtain $bu_{b,a}^2 \rightarrow y_a$. The finite run expansion may also be differentiated uniformly on this compact neighborhood, and its derivative tends to $H'(y_a) > 0$. Substitution of $u = c_a/\sqrt{b} + \alpha/b$ into (16) then forces $\alpha = -1 - y_a Q(y_a)/H'(y_a)$, and the uniform remainder gives (17). If $\ell \geq 2$, all terms in $Q(y)$ are positive for $y > 0$, proving the strict inequality. $\square$

If all ℓ rare colors occur once, then $H(y) = y^\ell$ and $y_a = 1$, giving the explicit specialization

$$u_{b,1,\dots,1} = b^{-1/2} - \frac{\ell+1}{2b} + O_\ell(b^{-3/2}).$$

An independent exact formula in this case is

$$S_{(b,1,\dots,1)}(u) = \ell! \sum_{q=1}^{\min(b,\ell+1)} \binom{\ell+1}{q} \binom{b-1}{q-1} u^{q+\ell-1}.$$

Indeed, order the ℓ distinct rare letters, choose the q gaps among their $\ell+1$ end and interior gaps that contain large-color runs, and choose the positive lengths of those runs. In particular two singleton rare colors give

$$u_{b,1,1} = b^{-1/2} - \frac{3}{2}b^{-1} + O(b^{-3/2}).$$

This agrees with the exact polynomial $S_{(b,1,1)}(u) = 6u^2 + 6(b-1)u^3 + (b-1)(b-2)u^4$. Theorem 4.6 is not uniform in the rare counts. Section 5 gives the leading crossing uniformly for total rare count $A = o(b)$.

5 Uniform completion of the near-vertex regime

For the leading term of the crossing, the fixed-count hypothesis in Theorem 4.6 can be removed, and the result is uniform over all rare counts with total $o(b)$. Proposition 5.5 shows that the corresponding approximation to the probabilities is not uniform. Its error can blow up while the relative error of the crossing tends to zero.

Throughout this section, $\ell \geq 1$ is fixed, $a_1, \dots, a_\ell$ are positive integers, $A = \sum_i a_i$, and

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad \kappa(y) = \frac{yH'(y)}{H(y)}.$$

Write y_a for the unique positive solution of $H(y_a) = 1$.

Lemma 5.1 (elasticity and concavity of the rare-count polynomial). *The polynomial H is convex, $\log H$ is concave on $(0, \infty)$, and κ is nondecreasing. Moreover,*

$$\frac{1}{A+1} \leq y_a \leq 1, \quad \kappa(y) \leq \ell + \sqrt{\ell A y}, \quad \kappa(y) \geq \max\{\ell, \tfrac{1}{2}\sqrt{A y}\} \quad (0 < y \leq 1). \quad (18)$$

Proof. Convexity follows directly from the nonnegative coefficients. For one factor, give $J \in \{0, \dots, a-1\}$ probabilities proportional to

$$r_j = \binom{a-1}{j} \frac{y^j}{(j+1)!}.$$

The successive ratio $R(j) = (j+1)r_{j+1}/r_j = y(a-1-j)/(j+2)$, with $R(a-1) = 0$, is nonincreasing. Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}R(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)R(J)]$. Therefore

$$\text{Var } J = \mathbb{E}J + \text{Cov}(J, R(J)) \leq \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J' , twice the covariance equals $\mathbb{E}[(J - J')(R(J) - R(J'))] \leq 0$. Putting $M = J + 1$ and $\kappa_a = yB'_a/B_a = \mathbb{E}M$, we obtain

$$(\log B_a)'' = \frac{\text{Var } M - \mathbb{E}M}{y^2} \leq 0, \quad y\kappa'_a = \text{Var } M \geq 0.$$

This proves the concavity of $\log H$ and the monotonicity of κ .

The elementary differential equation $yB''_a + yB'_a = aB_a$ gives

$$ay = \kappa_a^2 + \text{Var } M + (y-1)\kappa_a.$$

Using $0 \leq \text{Var } M \leq \kappa_a - 1$ yields $\kappa_a \leq 1 + \sqrt{ay}$ and $\kappa_a^2 + y\kappa_a \geq ay + 1$. For $0 < y \leq 1$, these imply $\kappa_a \geq \max\{1, \sqrt{ay} - 1/2\} \geq \frac{1}{2}\sqrt{ay}$. Summing over the factors and applying Cauchy–Schwarz proves the elasticity bounds. Finally, $B_a(1) \geq 1$ and

$$B_a(y) \leq y\Phi(ay) \leq ye^{ay/2},$$

where $(j+1)! \geq 2^j$ proves the last inequality term by term. Thus $H(1) \geq 1$, whereas $H(1/(A+1)) \leq (A+1)^{-\ell} e^{A/(2(A+1))} < 1$. $\square$

Lemma 5.2 (an exact compound-binomial representation). *For $0 < u < 1$, put $v = u/(1-u)$. Let G have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u . Take all these variables independent and set*

$$T = G + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^A \mathbb{E}[H(vT)], \quad \mathbb{E}e^{sT} = (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \quad (19)$$

In particular, set $y = bu^2$, $\kappa = \kappa(y)$ and $s = \kappa/[bu(1-u)]$. Whenever $s < 1$,

$$(1-u)^A H(y) \leq S_{(b,a)}(u), \quad (20)$$

$$S_{(b,a)}(u) \leq (1-u)^A H(y) e^{-\kappa} (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1}. \quad (21)$$

Proof. The exact positive-refresh integral is

$$S_{(b,a)}(u) = \frac{(1-u)^{b+A-1}}{v} \int_0^\infty e^{-t} B_b(vt) H(vt) dt.$$

The integral of $e^{-t} B_b(vt)$ is $v(1+v)^{b-1}$. After normalization its density is a mixture of gamma densities with shape $J+2$, where $J \sim \text{Bin}(b-1, u)$. This is exactly the distribution of T , proving the identity and its moment generating function. Since $\mathbb{E}T = 2 + (b-1)u$, $v\mathbb{E}T \geq bu^2 = y$. Convexity and monotonicity of H prove (20). Concavity of $\log H$ gives, for $w > 0$,

$$H(w) \leq H(y) \exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = vT$ and use the moment generating function to prove (21). $\square$

Theorem 5.3 (all growing rare counts near a vertex). *For fixed ℓ , uniformly over all positive rare counts with $A = o(b)$, the unique positive crossing odds satisfy*

$$u_{b,a} = \sqrt{\frac{y_a}{b}} \left(1 + O_\ell\left(\sqrt{\frac{A+1}{b}}\right)\right), \quad \prod_i B_{a_i}(y_a) = 1. \quad (22)$$

Explicitly, put

$$K = \ell + \sqrt{\ell} + 1, \quad \varepsilon = \sqrt{\frac{A+1}{b}}, \quad \delta = 16K\varepsilon.$$

If $\delta \leq 1/2$, then

$$(1-\delta)\sqrt{y_a/b} \leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}. \quad (23)$$

The constants are not optimized. No condition such as $A(\log A)^2 = o(b)$ is needed.

Proof. The proper-run polynomial has positive coefficients, zero constant coefficient, and is unbounded at infinity, so its crossing is unique. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since $\delta \leq 1/2$ forces $\varepsilon \leq 1/(32K)$ and $y_a \leq 1$, at $u_+ = (1+\delta)u_0$ we have $u_+ \leq \frac{3}{2}\varepsilon < 1/4$. Monotonicity of κ and the lower bound in Lemma 5.1 give

$$\log H(bu_+^2) \geq 2\kappa_0 \log(1+\delta), \quad \frac{Au_0}{\kappa_0} \leq 2\sqrt{A/b} \leq 2\varepsilon.$$

Consequently, by (20),

$$\log S(u_+) \geq 2\kappa_0 \log(1+\delta) - \frac{Au_+}{1-u_+} \geq \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

At $u_- = (1-\delta)u_0$, write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(A+1)$, $\delta \leq 1/2$, and (18) imply

$$u_- \leq \varepsilon, \quad bu_- \geq \frac{1}{2\varepsilon}, \quad s = \frac{\kappa}{bu_-(1-u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4K\varepsilon \leq \frac{1}{8}.$$

Using $\log(1+x) \leq x$, $-\log(1-s) \leq s/(1-s)$ and $u_-, s \leq 1/4$ in (21), and dropping the nonpositive term $A \log(1-u_-)$, gives

$$\begin{aligned} \log \frac{S(u_-)}{H(y)} &\leq -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\ &= \kappa \left(\frac{1}{(1-u_-)(1-s)} - 1 \right) + \frac{2s}{1-s} \\ &\leq 5\kappa(u_- + s) \leq 25K\kappa\varepsilon. \end{aligned}$$

Here $\kappa \geq \ell \geq 1$ was used in the last two inequalities. Again by monotonicity of κ ,

$$\log H(y) \leq 2\kappa \log(1 - \delta) \leq -2\kappa\delta.$$

Thus $\log S(u_-) \leq \kappa(-32K + 25K)\varepsilon < 0$. The two strict inequalities and uniqueness bracket the crossing. $\square$

Corollary 5.4 (an explicit shape law for growing rare counts). *Suppose $A \rightarrow \infty$, $A = o(b)$, and the actual proportions $\alpha_i = a_i/A$ stay in a fixed compact subset of the positive ℓ -simplex. Put $h = \sum_i \sqrt{\alpha_i}$. Then*

$$u_{b,a} \sim \frac{\ell \log A}{2h\sqrt{Ab}}.$$

More precisely, the polynomial root satisfies

$$2h\sqrt{Ay_a} = \ell \log A - \frac{\ell}{2} \log \log A + \frac{\ell}{2} \log \left(\frac{8\pi h}{\ell} \right) + \frac{3}{4} \sum_i \log \alpha_i + O_\ell \left(\frac{\log \log A}{\log A} + \frac{(\log A)^2}{A} \right).$$

Replacing $\sqrt{Ay_a}$ on the left by $\sqrt{Ab}u_{b,a}$ incurs an additional $O_\ell(\log A \sqrt{A/b})$ error.

Proof. Lemma 4.1, uniformly when $y \rightarrow 0$, gives $H(y) = y^\ell \prod_i \Phi(a_i y)(1 + O_\ell(y))$. Write $x = \sqrt{Ay}$. Bracketing first at fixed small and large multiples of $\log A$ shows that the root has $x \asymp \log A$. The standard large-argument expansion of I_1 then gives uniformly over the stated proportions

$$H(x^2/A) = \frac{x^{\ell/2} e^{2hx}}{A^\ell (4\pi)^{\ell/2} (\prod_i \alpha_i)^{3/4}} (1 + O_\ell(x^{-1} + x^2/A)).$$

Taking logarithms, substituting $x = (\ell/(2h)) \log A + O_\ell(\log \log A)$ in the logarithmic term, and using Theorem 5.3 proves the claims. The constants also depend on the fixed compact set of proportions. $\square$

Proposition 5.5 (probability approximation can fail while roots agree). *Let $a = \lfloor b/\log b \rfloor$, let y_a solve $B_a(y_a) = 1$, and set $u_0 = \sqrt{y_a/b}$. Then*

$$H(y_a) = B_a(y_a)^2 = 1, \quad S_{(b,a,a)}(u_0) \rightarrow \infty, \quad \frac{u_{b,a,a}}{u_0} \rightarrow 1.$$

Hence the crossing can be approximated uniformly even where $S(u) \sim H(bu^2)$ fails, so that approximation is not uniform over $A = o(b)$.

Proof. Retain only run words that start and end in the large color and have exactly one adjacent pair of distinct rare colors. The run enumeration in Theorem 4.6 shows that their contribution is the polynomial $2uyB'_a(y)^2$, with each term of total rare-run count m additionally multiplied by $\prod_{j=1}^{m-1} (1 - j/b)_+$. Give J probabilities proportional to $\binom{a-1}{j} y^j/j!$, the coefficients of $B'_a(y)$. Its differential equation $y(B'_a)'' + (1+y)(B'_a)' - (a-1)B'_a = 0$ implies $\mathbb{E}J^2 + y\mathbb{E}J = (a-1)y$, and hence $\mathbb{E}J \leq \sqrt{ay}$ and $\mathbb{E}J^2 \leq ay$. For two independent copies, $m = J_1 + J_2 + 2$, so

$$\mathbb{E}[m(m-1)] \leq 4ay + 6\sqrt{ay} + 2.$$

The clipped-product deficit bound therefore proves

$$S_{(b,a,a)}(u_0) \geq \frac{2\kappa_a(y_a)^2}{bu_0} \left(1 - \frac{4ay_a + 6\sqrt{ay_a} + 2}{2b} \right).$$

The preceding corollary for one rare color gives $\sqrt{ay_a} \sim \frac{1}{2} \log a$, and the elasticity bounds give $\kappa_a(y_a) \sim \sqrt{ay_a}$. The last display is asymptotic to a lower bound $\sqrt{a/b} \log a \sim \sqrt{\log b}$, which diverges. The root assertion follows directly from Theorem 5.3, since $2a = o(b)$. $\square$

6 Spatial superlevel sets and weak-field face phases

The interior-composition estimate also determines the geometry of the bins whose individual probabilities exceed the vertex probability. We state it with a weak external field, since the field produces face phases that do not appear in the symmetric window.

Fix d and the symmetric transition model of Section 3. For $h = (h_1, \dots, h_d) \in \mathbb{R}^d$, define the *tilted occupation law*

$$\tilde{P}_{N,h}(n) = \frac{e^{h \cdot n/N} P_N(n)}{\sum_{m_1 + \dots + m_d = N} e^{h \cdot m/N} P_N(m)}. \quad (24)$$

So each step feels a field of h_i/N . Note that we tilt the law of the counts, and the chain itself is unchanged. The normalizing sum cancels whenever we compare two bins. When we talk about a modal phase, we mean the face with the largest single bin probabilities, which need not be the face with most of the mass. Put $L = \log N$, keep

$$u_N(t) = \frac{L - \frac{1}{2} \log L + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)},$$

and write $h_* = \max_i h_i$ and $\tilde{V}_{N,h} = \max_i \tilde{P}_{N,h}(N e_i)$. For a support $S \subseteq \{1, \dots, d\}$ of cardinality $k \geq 2$, set

$$\bar{h}_S = \frac{1}{k} \sum_{i \in S} h_i, \quad B_S(t, h) = (k-1)(t - a_k) + \bar{h}_S - h_*, \quad H_S = \{y \in \mathbb{R}^S : \sum_{i \in S} y_i = 0\}.$$

Euclidean measure on H_S is understood, and $\omega_s = \pi^{s/2}/\Gamma(1 + s/2)$ denotes the volume of its s -dimensional unit ball when $s = k-1$.

Theorem 6.1 (Gaussian superlevel geometry). *For a count vector with support exactly S , put*

$$y_i = \sqrt{L} \left(\frac{n_i}{N} - \frac{1}{k} \right), \quad i \in S.$$

Uniformly when t, h, y range over fixed compact sets, with $y \in H_S$,

$$\frac{\tilde{P}_{N,h}(n)}{\tilde{V}_{N,h}} = \exp \left(B_S(t, h) - \frac{k^2}{4} \sum_{i \in S} y_i^2 + o(1) \right). \quad (25)$$

Let $M_{N,S}(t, h)$ be the number of bins of support exactly S whose tilted probability is strictly greater than $\tilde{V}_{N,h}$. Then, uniformly for (t, h) in compact sets,

$$\frac{L^{(k-1)/2}}{N^{k-1}} M_{N,S}(t, h) \longrightarrow \frac{\omega_{k-1}}{\sqrt{k}} \left(\frac{4[B_S(t, h)]_+}{k^2} \right)^{(k-1)/2}, \quad [x]_+ = \max(x, 0). \quad (26)$$

If $B_S(t, h) < 0$, there are no such bins for all sufficiently large N . At $B_S(t, h) = 0$ the normalized count tends to zero, but we do not claim that the set is eventually empty.

Proof. With $\alpha_i = 1/k + y_i/\sqrt{L}$ and $\sum_i y_i = 0$, Taylor expansion on any fixed compact set in H_S gives

$$\left(\sum_i \sqrt{\alpha_i} \right)^2 = k - \frac{k^2}{4L} \sum_i y_i^2 + O(L^{-3/2}), \quad D(\alpha) = D_k(1 + o(1)).$$

Insert these expressions and $z = L - \frac{1}{2} \log L + t$ into Theorem 3.5. Since $\log D_k = -(k-1)a_k$, the logarithm of the untilted ratio is

$$(k-1)(t - a_k) - \frac{k^2}{4} \sum_i y_i^2 + o(1).$$

The logarithm of the field factor relative to the largest vertex is $h \cdot \alpha - h_* = \bar{h}_S - h_* + o(1)$, uniformly on the stated compact sets. This proves (25).

It remains to rule out superlevel bins near the boundary of the face, where the local estimate does not apply. Fix compact sets of t, h , and choose a sufficiently large constant R . Starting from any positive count vector on S , successively transfer one unit from a coordinate b to a coordinate a when $b \geq a + 2$. These transfers preserve the support and terminate at a balanced vector. By Theorem 3.6, the untilted probability strictly increases at every transfer. Each transfer changes the scaled vector y by norm $\sqrt{2L}/N$ and decreases its squared norm. If the starting vector has $\|y\| > R$, the first vector along this path with norm at most R has norm $R + o(1)$, so it lies in a compact subset of the relative interior of the face. Its untilted log ratio to a vertex is at most

$$(k-1)(t - a_k) - k^2 R^2 / 4 + o(1).$$

This is an upper bound for the starting vector's untilted log ratio. Since $h \cdot \alpha - h_* \leq 0$ for every composition α , we can pick one R that works for all the parameters, so that every tilted superlevel bin has $\|y\| \leq R$. The same argument, with R larger if needed, also bounds where the maximum on a face can be, by comparing with the balanced bin of that face.

Inside this ball the logarithms in (25) converge uniformly. For any $\delta > 0$, their positive set is sandwiched between the lattice points in balls of squared radii $4[B_S - \delta]_+/k^2$ and $4[B_S + \delta]_+/k^2$, with the inner ball interpreted as empty when $B_S \leq \delta$. The scaled count vectors form a translate of the lattice

$$\frac{\sqrt{L}}{N} \{z \in \mathbb{Z}^S : \sum_i z_i = 0\} \quad \text{in } H_S.$$

Its covolume is $\sqrt{k}(\sqrt{L}/N)^{k-1}$, since the basis $e_i - e_k$, $1 \leq i < k$, has Gram determinant k . Counting lattice points in the two balls then gives the volume constant, and letting $\delta \downarrow 0$ gives (26), also when $B_S = 0$. On compact parameter sets the radii are bounded, so the convergence is uniform. Finally, if $B_S < 0$, the profile is below one on the whole ball for large N , so there are no such bins. $\square$

At zero field, all supports of the same size have the same count, so the right-hand side of (26) is multiplied by $\binom{d}{k}$ when counting all bins with k positive coordinates. For $k = 2$, writing $y = (y_1, -y_1)$ gives the profile $\exp(t - a_2 - 2y_1^2)$ and the one-face count constant $\sqrt{2[t - a_2]_+}$, which agrees with the free binary window once the different centering is taken into account.

Without a field, a proper face never wins in the window, because the full center beats every proper face as soon as it beats the vertex (Corollary 3.7). A field changes this. If some states are unfavorable, a face that avoids them can come out ahead for a range of t , and the next theorem says which face wins.

Theorem 6.2 (weak-field modal phase diagram). *Define the face scores*

$$\lambda_S(t, h) = \begin{cases} h_i, & S = \{i\}, \\ (|S| - 1)(t - a_{|S|}) + \bar{h}_S, & |S| \geq 2. \end{cases}$$

The largest bin weight, expressed relative to the untilted vertex before normalizing (24), satisfies

$$\log \max_n \left\{ e^{h \cdot n/N} \frac{P_N(n)}{V_N} \right\} \longrightarrow \max_{\emptyset \neq S \subseteq \{1, \dots, d\}} \lambda_S(t, h). \quad (27)$$

For fixed (t, h) , every global mode has, for all sufficiently large N , a support maximizing the limiting score. If the maximizing support is unique, all global modes have exactly that support. On any selected support S with $|S| \geq 2$, their locations satisfy

$$\max_{i \in S} \left| \frac{n_i}{N} - \frac{1}{|S|} \right| = o((\log N)^{-1/2}).$$

This does not identify the maximizing vectors exactly, and equal limiting scores do not imply that both supports hold modes at finite N .

If the fields are sorted $h_1 \geq \dots \geq h_d$, the score maximization reduces to the upper envelope of the d lines

$$\lambda_1(t) = h_1, \quad \lambda_k(t) = (k-1)(t - a_k) + \frac{1}{k} \sum_{i=1}^k h_i, \quad 2 \leq k \leq d. \quad (28)$$

Away from ties, the winning support size is nondecreasing as t increases. If the fields are strictly ordered, the corresponding supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d-1\}$ can be the unique winning size on an open interval for suitable fixed fields. Thus proper-face phases can persist asymptotically under a field of order $1/N$ per step.

Proof. For each fixed support, the confinement argument in the preceding proof applies also to its maximum, because the field factor is bounded above and below uniformly in N . On that compact scaled region, the profile relative to the untilted vertex is $\exp(\lambda_S - k^2 \|y\|^2/4 + o(1))$. Balanced integer vectors have $y = o(1)$, so the face maximum converges to e^{λ_S} . There are finitely many supports. Taking their maximum proves (27), and the positive gap to any nonmaximizing score excludes its support for large N . If a sequence of maximizers within a selected face had scaled norm bounded away from zero, the uniform Gaussian profile would make it strictly smaller than that face's balanced bin. This proves the location statement.

Among supports of a given size k , the largest field average is the average of the k largest fields. This proves (28). For $t_1 < t_2$, a line of strictly larger slope cannot win at t_1 and lose to a line of smaller slope at t_2 , because subtracting the two comparisons would give $(k_1 - k_2)(t_2 - t_1) \leq 0$. Hence the winning size is nondecreasing away from ties.

To realize a prescribed $2 \leq k < d$, set $h_1 = \dots = h_k = 0$ and $h_{k+1} = \dots = h_d = -H$. Choose $t > a_k$ sufficiently close to a_k so that $t < a_{k-1}$ if $k > 2$. Then the k -face score is positive, all smaller nonvertex face scores are negative, and the largest vertex score is zero. For $j > k$, the largest score is

$$(j-1)(t - a_j) - H \frac{j-k}{j}.$$

Taking H sufficiently large makes each of these strictly smaller than the k -face score. Every other support of size k includes a negative field and also loses. All comparisons are strict, so they persist on an open interval of t . Small perturbations can make the fields strictly ordered without removing the phase. $\square$

Corollary 6.3 (one field realizes the full face hierarchy). *Fix $d \geq 2$ and $H > 3/4$, and choose*

$$h_i = -H(i-1)^2, \quad 1 \leq i \leq d.$$

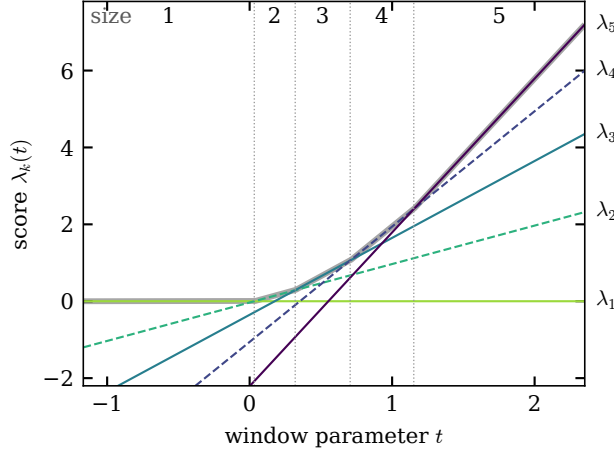


Figure 2: The lines (28) for $d = 5$ and $h_i = -(i-1)^2$. The grey band is their upper envelope. The dotted lines are $\tau_1 < \dots < \tau_4$, and the number at the top of each interval is the size k of the winning support $\{1, \dots, k\}$.

In particular $h_i = -(i-1)^2$ is a single explicit choice valid in every fixed dimension. Put

$$c_k = (k + \frac{1}{2}) \log k - \frac{k-1}{2} \log(4\pi), \quad \tau_k = c_k - c_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and set $\tau_0 = -\infty$, $\tau_d = +\infty$. Then $\tau_1 < \dots < \tau_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support throughout the nonempty interval $\tau_{k-1} < t < \tau_k$. Consequently, for each fixed t in this interval and all sufficiently large N , every global mode of the tilted law has exactly those k positive coordinates.

Proof. Here $c_1 = 0$ and $c_k = -(k-1)a_k$ for $k \geq 2$. Since $k^{-1} \sum_{i=1}^k (i-1)^2 = (k-1)(2k-1)/6$, the lines in (28) have intercepts $c_k - H(k-1)(2k-1)/6$. The intersection of lines k and $k+1$ is therefore $t = \tau_k$. Extend c_k to the displayed smooth function $c(x)$ for $x \geq 1$. Direct differentiation gives

$$0 < c''(x) = \frac{x-1/2}{x^2} \leq \frac{1}{2}.$$

Integrating this bound twice over unit intervals yields $c_{k+2} - 2c_{k+1} + c_k \leq 1/2$. Hence

$$\tau_{k+1} - \tau_k = \frac{2H}{3} - (c_{k+2} - 2c_{k+1} + c_k) > 0.$$

For $\tau_{k-1} < t < \tau_k$, summing consecutive differences $\lambda_{j+1}(t) - \lambda_j(t) = t - \tau_j$ shows that line k strictly exceeds every earlier and later line. The fields are strictly ordered, so its unique best support is $\{1, \dots, k\}$. Apply Theorem 6.2 to obtain the finite- N conclusion. $\square$

Figure 2 shows the lines of Corollary 6.3 for $d = 5$.

Example 6.4 (a persistent triangular edge phase). For $d = 3$ and $h = (0, 0, -H)$, the three competing maximal scores are

$$0, \quad t - a_2, \quad 2(t - a_3) - H/3.$$

If $H > 6(a_2 - a_3)$, the edge with support $\{1, 2\}$ wins strictly for

$$a_2 < t < H/3 + 2a_3 - a_2.$$

For every fixed t in this interval, all global modes of the tilted law lie on that edge for sufficiently large N . Without the field, no proper face holds a mode in the critical window (Corollary 3.7).

7 Verification and open questions

The proofs above do not rely on computation. The Python scripts for this paper use only the standard library and check the results at finite size. `verify_geometry.py` compares 6,733 exact rational probabilities in 237 distributions with a dynamic program, and 158 of the distributions also with exhaustive enumeration. It checks normalization, means, covariances, and the values at $p = 0, 1$ for $d = 2, 3, 4$. `verify_simplex_threshold.py` checks the positive refresh formula, the balancing transfers, the finite-size edge mode, and the crossing expansions. `verify_frontier_boundary.py` and `verify_frontier_modes.py` check the uniform comparison, the rare-coordinate formulas, the spatial profiles, and the weak-field results against finite calculations. Finally, `verify_vertex_completion.py` makes 462 exact identity comparisons, 200 exact checks of the elasticity bounds, 16 root-bracket checks, and 4 large cases with growing rare counts. Everything they print is saved in `data/`. The decimal checks are only numerical, and none of the proofs rely on them.

Some things are still open. We know the leading near-vertex crossing for all of $A = o(b)$, but not the next term uniformly in that range. We also have not looked at a growing number of colors, at switching that is not uniform, or at the full mode diagram for finite N outside the critical window. Example 3.8 and Proposition 5.5 show that the simplest strengthenings of our results are false.

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